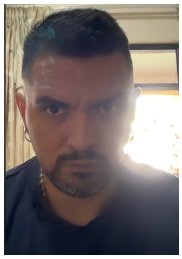
# Omar Corona Tejeda



**AI TRAINER & EVALUATOR | MATHEMATICAL REASONING & FORMAL VERIFICATION**

Mexico City, Mexico | (+52) 272-283-9135
omarct1989@gmail.com | omarct1989@ciencias.unam.mx
LinkedIn | Mathematical Physics Project

AI and data evaluator with experience assessing speech naturalness, voice similarity, audio quality, model responses, and search relevance. Develops advanced mathematical problems, evaluation rubrics, and reference solutions to test the reasoning, accuracy, and reliability of artificial intelligence models, and formalizes proofs using Lean 4 and Mathlib. Brings together experience in multimodal data annotation, quality assurance, and university-level teaching, grounded in an academic background in mathematics and physics.

## CORE EXPERTISE

**Mathematical Reasoning & Formal Verification:** Advanced problem design, proof analysis, Lean 4 & Mathlib.

**AI Training & Evaluation:** Response assessment, rubrics, reference solutions, multimodal data quality.

**Teaching & Technical Communication:** University-level instruction, mathematical exposition, technical writing.

## AI & DATA EXPERIENCE

### AI & Data Quality Evaluator | TELUS | 2026

- Evaluates speech and generated audio for voice similarity, naturalness, pronunciation, clarity, and audio quality; compares outputs against reference recordings and documents rating justifications.
- Assesses search results for relevance and usefulness against user intent; applies rating guidelines and writes comparative justifications.

### Handshake | 2026

- Evaluates mathematical responses and proofs using structured rubrics for correctness, completeness, and logical consistency; documents errors and gaps in reasoning.
- Develops and reviews advanced mathematics prompts, assessment rubrics, and reference solutions, including problems in analysis and mathematical physics.
- Formalizes mathematical arguments in Lean 4 with Mathlib; checks project builds, theorem assumptions, and proof dependencies to support reproducible verification.

### AI Data Annotator (Spanish - Mexico) | RWS | 2026

- Evaluates and tags Spanish (Mexico) data to improve AI assistant response quality.
- Reviews voice conversations for naturalness, turn-taking, interruptions, and audio quality; documents specific errors and assesses whether models correct them after feedback.
- Annotates and transcribes audio, video, images, and text; contributes recordings and written inputs and outputs for multimodal training datasets.

### Robotics Data Collector - Quality Assurance | Remotasks | 2025-2026

- Collected robotics data and performed quality assurance for generative AI model projects.

### AI Trainer | Outlier AI | 2024-2025

- Contributed to AI training projects requiring structured analytical and written work.

## EDUCATION & PROFESSIONAL TRAINING

- 2008-2013: Bachelor's Degree in Mathematical Sciences, Faculty of Sciences, UNAM. Professional License: 14085315.
- 2014-2017: Master's Degree in Mathematical Sciences, Institute of Mathematics, UNAM. Professional License: 14030555.
- 2018-present: PhD in Mathematical Sciences (in progress), Institute of Mathematics, UNAM.
- 2023-present: Bachelor's Degree in Physics (in progress), Faculty of Sciences, UNAM.

## PROFESSIONAL PROJECTS

**The Mathematical Physics Project | Creator & Curator**

- Created and maintains a growing digital library of self-contained notes, books, exercises, and reference materials for students, instructors, and researchers.
- Publishes and organizes downloadable resources across quantum theory, relativity, gauge theory, geometry, algebra, operator algebras, and classical mechanics.

## TEACHING EXPERIENCE

- 2024: University Algebra I, Tiffin University (online).
- Differential and Integral Calculus, Southern New Hampshire University (online).
- Discrete Mathematics, Saint Leo University (online).
- 2023: Control Theory, General Relativity, Quantum Mechanics, Theoretical Mechanics, and Quantum Field Theory.
- 2019-2020: Riemannian Geometry, Master's Program in Mathematical Sciences.
- 2016-2017: Mathematics I, II, III, Instituto Nueva Inglaterra (secondary level).
- 2014-2016: Adjunct Professor, Linear Algebra I, Faculty of Sciences, UNAM.
- 2008-2009: Mathematics Instructor for Undergraduate Admissions, Faculty of Sciences, UNAM.

## RESEARCH AFFILIATIONS

- Center of Mathematical Sciences and Applications (CMSA), Harvard University, Cambridge, Massachusetts.
- Max Planck Institute for Mathematics, Bonn, Germany.

## SELECTED RESEARCH ACTIVITIES

- 2025: Participant, ICMAM Latin America Satellite Conference on Differential Geometry.
- Participant, Twinned Workshop on Quantum Field Theory and Topological Phases via Homotopy Theory and Operator Algebras, Max Planck Institute for Mathematics and CMSA.
- Presenter, Category Theory Seminar, Institute of Mathematics, UNAM.
- Member, PAPIIT Research Project: Structures in Tricategories: Monads and their Generalizations (IN110111).

## LANGUAGES & TECHNICAL SKILLS

**Languages:** Spanish: Native | English: Advanced | French: Intermediate

**Formal Verification:** Lean 4, Mathlib, Lake; mathematical formalization, proof checking, and axiom dependency review.

**Programming & Scientific Computing:** Python, Java, MATLAB, Simulink, Anaconda.

**Development Tools & Version Control:** Visual Studio Code, Xcode, Git, GitHub, terminal/shell; project builds, script execution, and verification workflows.

**Application & Web Development:** Swift, SwiftUI, HTML, CSS, JavaScript, React, Vite; project experience with AI-assisted development.

**Technical Writing & Publishing:** LaTeX; preparation of mathematical documents and maintenance of a digital resource library with GitHub Pages.

**AI-Assisted Workflows:** Codex; assisted coding, debugging, mathematical formalization, and document preparation.